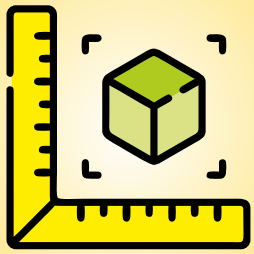
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PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Unit & Dimension, Basic Maths and Vectors

(English Medium)

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UNIT & DIMENSION, BASIC MATHS AND VECTOR

KEY CONCEPT

UNIT & DIMENSION

The following is the list of some physical quantities with their formulae and dimensional formulae with units

S.No.	Physical Quantity	Explanation or Formulae	Dimensional Formulae	S.I. Unit
1.	Distance, Displacement, Wave length, Radius of gyration, Circumference, Perimeter, Light year		$[M^0 L^1 T^0]$	m
2.	Mass		$[M^1 L^0 T^0]$	kg
3.	Period of oscillation,	$\frac{\text{total time}}{\text{no. of oscillations}}$		
	Time,		$[M^0 L^0 T^1]$	s
4.	Frequency	Reciprocal of time period $n = \frac{1}{T}$	$[M^0 L^0 T^{-1}]$	herts (Hz)
5.	Area	A = length × breadth	$[M^0 L^2 T^0]$	m^2
6.	Volume	V = length × breadth × height	$[M^0 L^3 T^0]$	m^3
7.	Density	$d = \frac{\text{mass}}{\text{volume}}$	$[M^1 L^{-3} T^0]$	kgm^{-3}
8.	Linear mass density	$\lambda = \frac{\text{mass}}{\text{length}}$	$[M^1 L^{-1} T^0]$	kgm^{-1}
9.	Speed, Velocity	$v = \frac{\text{displacement}}{\text{time}}$	$[M^0 L^1 T^{-1}]$	ms^{-1}
10.	Acceleration	$a = \frac{\text{change in velocity}}{\text{time}}$	$[M^0 L^1 T^{-2}]$	ms^{-2}
11.	Linear momentum	P = mass × velocity	$[M^1 L^1 T^{-1}]$	$kgms^{-1}$
12.	Force	F = Mass × acceleration	$[M^1 L^1 T^{-2}]$	N
13.	Impulse	J = Force × time	$[M^1 L^1 T^{-1}]$	Ns
14.	Work, Energy, PE, KE, Strain energy,	W = Force × displacement PE = mgh		
	Heat energy	$KE = \frac{1}{2}$ (Mass) (Velocity)^2	$[M^1 L^2 T^{-2}]$	J (or) N.m
15.	Power	$P = \frac{\text{Work}}{\text{Time}}$	$[M^1 L^2 T^{-3}]$	Watt

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16.	Pressure, Stress	$\frac{\text{Force}}{\text{Area}}$		
Modulus of elasticity (Y, η, k)		$Y = \frac{\text{Stress}}{\text{Strain}}$	$[M^1 L^{-1} T^{-2}]$	Pascal or Nm^{-2}
17.	Strain	$= \frac{\text{change in dimesion}}{\text{original dimension}}$	$[M^0 L^0 T^0]$	no units
18.	Strain energy density	$E = \frac{\text{work}}{\text{volume}}$	$[M^1 L^{-1} T^{-2}]$	Jm^{-3}
19.	Angular displacment	$\theta = \frac{\text{length of arc}}{\text{radius}}$	$[M^0 L^0 T^0]$	rad
20.	Angular velocity	$\omega = \frac{\text{angular displacement}}{\text{time}}$	$[M^0 L^0 T^{-1}]$	$rads^{-1}$
21.	Angular acceleration	$\alpha = \frac{\text{chang in angular velocity}}{\text{time}}$	$[M^0 L^0 T^{-2}]$	$rads^{-2}$
22.	Angular momentum	L = linear momentum × perpenducular distance	$[M^1 L^2 T^{-1}]$	Js
23.	Planck's constant	$h = \frac{\text{energy}}{\text{frequency}}$	$[M^1 L^2 T^{-1}]$	Js
24.	Angular impulse	Torque × time	$[M^1 L^2 T^{-1}]$	Js
25.	Torque	τ = force × ⊥ distance	$[M^1 L^2 T^{-2}]$	Nm
26.	Acceleration due to gravity(g)	$g = \frac{\text{weight}}{\text{mass}}$	$[M^0 LT^{-2}]$	ms^{-2} or Nkg^{-1}
27.	Universal gravitational Constant	$G = \frac{\text{Force} \times (\text{distance})^2}{\text{Mass}_1 \times \text{Mass}_2}$	$[M^{-1}L^3 T^{-2}]$	$Nm^2 kg^{-2}$
28.	Moment of inertia	I = Mass × (radius of gyration)^2	$[M^1 L^2 T^0]$	kgm^2
29.	Velocity gradient	$= \frac{dv}{dx}$	$[M^0 L^0 T^{-1}]$	s^{-1}
30.	Surface tension, Spring constant Force constant	$S = \frac{\text{suface energy}}{\text{change in area}} = \frac{\text{force}}{\text{length}}$ $S = \frac{\text{force}}{\text{elongation}}$	$[M^1 L^0 T^{-2}]$	Nm^{-1} or Jm^{-2}
31.	Coefficienty of viscosity	$\eta = \frac{\text{tangential stress}}{\text{velocity gradient}}$	$[M^1 L^{-1} T^{-1}]$	Pa s (or) $Nm^{-2}s$
32.	Gravitational potential	Gravitational field × distance	$[M^0 L^2 T^{-2}]$	J/Kg
33.	Heat energy	msθ	$[M^1 L^2 T^{-2}]$	Joule
34.	Temperature	θ	$[M^0 L^0 T^0 \theta^1]$	kelvin (K)

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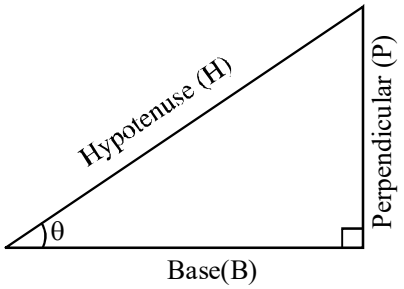
35.	Specific heat capacity	$S = \frac{\text{heat energy}}{\text{mass} \times \text{temp.}}$	$[M^0 L^2 T^{-2} \theta^{-1}]$	$Jkg^{-1} K^{-1}$
36.	Thermal capacity	$\frac{dQ}{d\theta}$ = mass × specific heat	$[M^1 L^2 T^{-2} \theta^{-1}]$	JK^{-1}
37.	Latent heat (or)			
	Calorific value	$L = \frac{\text{heat energy}}{\text{mass}}$	$[M^0 L^2 T^{-2}]$	Jkg^{-1}
38.	Water equivalent	W = Mass × specific heat	$[M^1 L^0 T^0]$	kg
39.	Coefficient of thermal expansion	$\alpha = \frac{\Delta\ell}{\ell\Delta\theta}$; $\beta = \frac{\Delta A}{A\Delta\theta}$; $\gamma = \frac{\Delta V}{V\Delta\theta}$	$[M^0 L^0 T^0 \theta^{-1}]$	K^{-1}
40.	Universal gas constant	$R = \frac{PV}{nT}$	$[M^1L^2T^{-2} \theta^{-1}mol^{-1}]$	$Jmol^{-1}K^{-1}$
41.	Boltzmann's constant			
	(for 1 Molecule)	$k = \frac{R}{\text{Avagadro number}}$	$[M^1L^2T^{-2} \theta^{-1}]$	$JK^{-1}molecule^{-1}$
42.	Mechanical equivalent of heat	$J = \frac{W}{H}$	$[M^0L^0T^0]$	no. SI units
43.	Coefficient of thermal conductity	$K = \frac{Qd}{A\,\Delta\theta t}$	$[M^1L^1T^{-3}\theta^{-1}]$	$Js^{-1}m^{-1}K^{-1}$ (or) $Wm^{-1}K^{-1}$
44.	Entropy	$\frac{dQ}{T} = \frac{\text{heat energy}}{\text{temperature}}$	$[M^1L^2T^{-2}\theta^{-1}]$	JK^{-1}
45.	Stefan's constant conductity	$\sigma = \frac{\Delta E}{\Delta A \Delta T \theta^4}$	$[M^1L^0T^{-3}\theta^{-4}]$	$Js^{-1}m^{-2}K^{-4}$ (or) $Wm^{-2}K^{-4}$
46.	Thermal resistance	$R = \frac{d\theta}{\left(\frac{dQ}{dt}\right)} = \frac{\text{temp} \times \text{time}}{\text{Heat}}$	$[M^{-1}L^{-2}T^3\theta^1]$	KsJ^{-1}
		(or) $R = \frac{d}{KA}$		
47.	Temperature gradient	$\frac{\text{Change in temperature}}{\text{length}} = \frac{d\theta}{dl}$	$[M^0L^{-1}T^0\theta^1]$	Km^{-1}
48.	Pressure gradient	$\frac{\text{Change in pressure}}{\text{length}} = \frac{dp}{dl}$	$[M^1L^{-2}T^{-2}]$	pascal m^{-1}
49.	Solar constant	$\frac{\text{energy}}{\text{area} \times \text{time}} = \frac{\Delta E}{AT}$	$[M^1L^0T^{-3}]$	$Js^{-1}m^{-2}$ (or) Wm^2

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BASIC MATHS

TRIGONOMETRIC RATIOS (T-RATIOS)

Following ratios of the sides of a right angled triangle are known as trigonometrical ratios.



$\sin\theta = \frac{P}{H}$ $\cos\theta = \frac{B}{H}$ $\tan\theta = \frac{P}{B}$ $\text{cosec}\theta = \frac{1}{\sin\theta} = \frac{H}{P}$

$\sec\theta = \frac{1}{\cos\theta} = \frac{H}{B}$ $\cot\theta = \frac{1}{\tan\theta} = \frac{B}{P}$

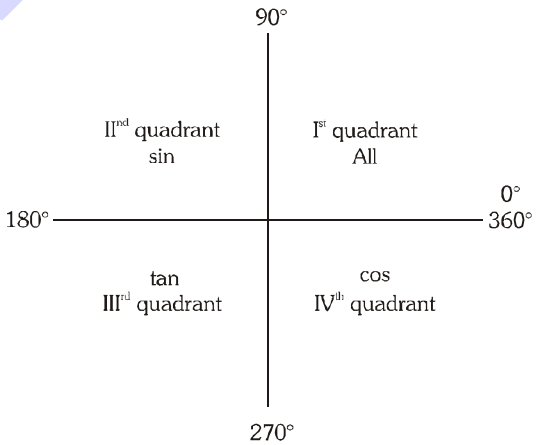
Trigonometric Identities

Commonly Used Values of Trigonometric Functions

Angle(θ)	0°	30°	37°	45°	53°	60°	90°
sin θ	0	$\frac{1}{2}$	$\frac{3}{5}$	$\frac{1}{\sqrt{2}}$	$\frac{4}{5}$	$\frac{\sqrt{3}}{2}$	1
cos θ	1	$\frac{\sqrt{3}}{2}$	$\frac{4}{5}$	$\frac{1}{\sqrt{2}}$	$\frac{3}{5}$	$\frac{1}{2}$	0
tan θ	0	$\frac{1}{\sqrt{3}}$	$\frac{3}{4}$	1	$\frac{4}{3}$	$\sqrt{3}$	∞

Four Quadrants and ASTC Rule

In first quadrant, all trigonometric ratios are positive.
In second quadrant, only sinθ and cosecθ are positive.
In third quadrant, only tanθ and cotθ are positive.
In fourth quadrant, only cosθ and secθ are positive



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Trigonometrical Ratios of General Angles (Reduction Formulae)

(i) Trigonometric function of an angle $2n\pi + \theta$ where n=0, 1, 2, 3,.... will remain same.

$\sin(2n\pi+\theta) = \sin\theta$ $\cos(2n\pi+\theta) = \cos\theta$ $\tan(2n\pi+\theta)=\tan\theta$

(ii) Trigonometric function of an angle $\left(\frac{n\pi}{2}+\theta\right)$ will remain same if n is even and sign of trigonometric function will be according to value of that function in quadrant.

$\sin(\pi-\theta) = +\sin\theta$ $\cos(\pi-\theta) = -\cos\theta$ $\tan(\pi-\theta)= -\tan\theta$

$\sin(\pi+\theta)= -\sin\theta$ $\cos(\pi+\theta)= -\cos\theta$ $\tan(\pi+\theta)= +\tan\theta$

$\sin(2\pi-\theta)= -\sin\theta$ $\cos(2\pi-\theta)= +\cos\theta$ $\tan(2\pi-\theta)= -\tan\theta$

(iii) Trigonometric function of an angle $\left(\frac{n\pi}{2}+\theta\right)$ will be changed into co-function if n is odd and sign of trigonometric function will be according to value of that function in quadrant.

$$\sin\left(\frac{\pi}{2}+\theta\right) = +\cos\theta \qquad \cos\left(\frac{\pi}{2}+\theta\right) = -\sin\theta \qquad \tan\left(\frac{\pi}{2}+\theta\right) = -\cot\theta$$

$$\sin\left(\frac{\pi}{2}-\theta\right) = +\cos\theta \qquad \cos\left(\frac{\pi}{2}-\theta\right) = +\sin\theta \qquad \tan\left(\frac{\pi}{2}-\theta\right) = +\cot\theta$$

(iv) Trigonometric function of an angle $-\theta$ (negative angles)

$\sin(-\theta)=-\sin\theta$ $\cos(-\theta)=+\cos\theta$ $\tan(-\theta)=-\tan\theta$

Addition/Subtraction Formulae for Trigonometrical Ratios

- $\sin(A+B)=\sin A\cos B + \cos A\sin B$
- $\sin(A-B)=\sin A\cos B-\cos A\sin B$
- $\cos(A+B)=\cos A\cos B-\sin A\sin B$
- $\cos(A-B)=\cos A\cos B + \sin A\sin B$

Small Angle Approximation

If θ is small (say $< 10°$) then $\sin\theta \approx \theta$, $\cos\theta \approx 1$ and $\tan\theta \approx \theta$

Note : here θ must be in radian.

BINOMIAL EXPRESSION

An algebraic expression containing two terms is called a binomial expression.

For example $(a+b)$, $(a+b)^3$, $(2x-3y)^{-1}$, $\left(x+\frac{1}{y}\right)$ etc. are binomial expressions.

Binomial Theorem

$$(a+b)^n = a^n + na^{n-1}b^1 + \frac{n(n-1)}{2\times 1}a^{n-2}b^2 + ... \qquad (1+x)^n = 1 + nx + \frac{n(n-1)}{2\times 1}x^2 + ...$$

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Binomial Approximation

If x is very small, then terms containing higher powers of x can be neglected so

$(1+x)^n = 1 + nx$

Greek Alphabet

A, α – alpha	I, ι – iota	P, ρ – rho
B, β – beta	K, κ – kappa	Σ, σ – sigma
Γ, γ – gamma	Λ, λ – lambda	T, τ – tau
Δ, δ – delta	M, μ – mu	Y, υ – upsilon
E, ε – epsilon	N, ν – nu	Φ, ϕ – phi
Z, ζ – zeta	Ξ, ξ – xi	X, χ – chi
H, η – eta	O, ο – omicron	Ψ, ψ – psi
Θ, θ – theta	Π, π – pi	Ω, ω – omega

Numerical Constants and Approximations

Numerical constants	Approximate formula (for $\alpha << 1$)
$\pi = 3.1416$	$(1 \pm \alpha)^n \approx 1 \pm n\alpha$
$\pi^2 = 9.8696$	$e^{\alpha} \approx 1 + \alpha$
$\sqrt{\pi} = 1.7725$	$\ln (1 + \alpha) \approx \alpha$
$e = 2.7183$	$\sin \alpha \approx \alpha$
$\log e = 0.4343$	$\cos \alpha \approx 1 - \alpha^2/2$
$\ln 10 = 2.3026$	$\tan \alpha \approx \alpha$

VECTORS

Precise description of laws of physics and physical phenomena requires expressing them in form of mathematical equations. In doing so we encounter several physical quantities, some of them have only magnitude and other have direction in addition to magnitude. Quantities of the former kind are referred as scalars and the latter as vectors and mathematical operations with vectors are collectively known as vector analysis.

VECTORS

A vector has both magnitude and sense of direction, and follows triangle law of vector addition. For example, displacement, velocity, and force are vectors.

Vector quantities are usually denoted by putting an arrow over the corresponding letter, as $\vec{A}$ or $\vec{a}$. Sometimes in print work (books) vector quantities are usually denoted by boldface letters as **A** or **a**.

Magnitude of a vector $\vec{A}$ is a positive scalar and written as $|\vec{A}|$ or A.

Unit Vector

It is mathematical way to express direction of a vector and defined by the ratio of a vector to its magnitude. When a unit vector is multiplied with a scalar magnitude, we get a vector of corresponding magnitude in the direction of the unit vector. A unit vector is usually represented by putting a sign (^) known as cap, hat or caret over a letter assigned to the unit vector. This letter may be the same as used for the vector, or its lower case letter, or some other symbol. For example, if we assign lower case letter a to unit vector in the direction of vector $\vec{A}$, the unit vector denoted by $\hat{a}$ is expressed by the following equation.

$$\vec{A} = A\hat{a}$$

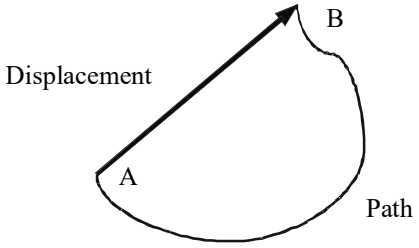
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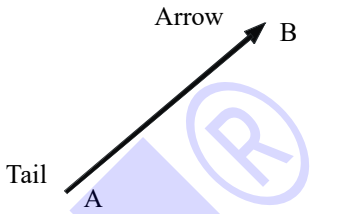
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Geometrical Representation of Vectors

Geometrically a vector is represented by a directed straight-line segment drawn to a scale. Starting point of the directed line segment is known as tail and the end-point as arrow, head, or tip. The orientation of the line and the arrow collectively show the direction and the length of the line drawn to a scale shows the magnitude.

For example let a particle moves from point A to B following a curvilinear path shown in the figure. It displacement vector is straight line AB directed form A to B. If straight-line distance between A and B is 25 m, the directed line segment has to be drawn to suitable scale. If we assume the scale 1.0 cm = 10 m, the geometrical length of the displacement vector AB must be 2.5 cm.





Geometrical representation of Displacement Vector

Addition of Vectors: The Triangle Law

Use of geometry in solving problems involving vectors is of fundamental nature. The triangle law also uses principles of plane geometry. This law states:

The vectors to be added are drawn in such a manner that the tail of a vector coincides the tip of the preceding vector (in tip to tail fashion); their resultant is defined by the vector drawn from the tail of the first vector to the tip of the second vector. The two vectors to be added and their resultant are coplanar.

Consider vectors $\vec{A}$ and $\vec{B}$ shown in the figure-I. Using the triangle law, we obtain geometrical construction shown in the figure-II, where it is shown that two vectors and their sum $\vec{C} = \vec{A} + \vec{B}$ always make a closed triangle. If we change order of vector $\vec{A}$ and $\vec{B}$, it shown in figure-III that sum given by equation $\vec{C} = \vec{B} + \vec{A}$ remain unchanged. Therefore, vector addition is commutative.

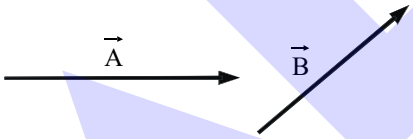


Fig. (i)

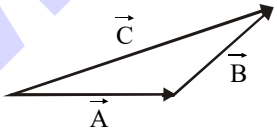


Fig. (ii)

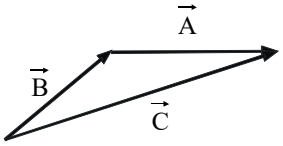
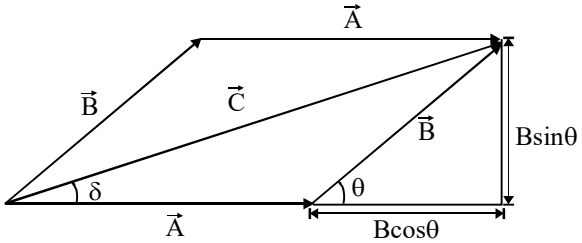


Fig. (iii)

Construction, which is combination of the figure-II and III, is in form of a parallelogram and is shown in figure-IV.



Geometry of the above figure suggests the following results.

$$C = \sqrt{A^2 + B^2 + 2AB\cos\theta}\ ;\quad \delta = \tan^{-1}\left(\frac{B\sin\theta}{A + B\cos\theta}\right)$$

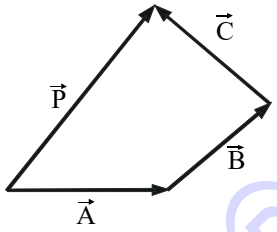
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Addition of more than two Vectors

The triangle law can be extended to define addition of more than two vectors. Accordingly, if vectors to be added are drawn in tip to tail fashion, resultant is defined by a vector drawn from the tail of the first vector to the tip of the last vector. This is also known as the polygon rule for vector addition.

Operation of addition of three vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ and their resultant $\vec{P}$ are shown in figure.

$$\vec{A}+\vec{B}+\vec{C}=\vec{P}$$

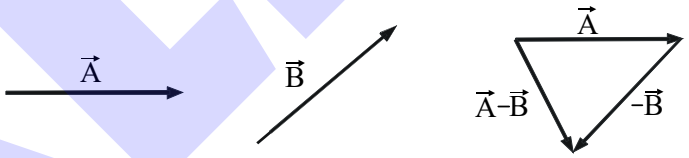


Here it is not necessary that three or more vectors and their resultant are coplanar. In fact, the vectors to be added and their resultant may be in different planes. However if all the vectors to be added are coplanar, their resultant must also be in the same plane containing the vectors.

Subtraction of Vectors

A vector opposite in direction but equal in magnitude to another vector $\vec{A}$ is known as negative vector of $\vec{A}$. It is written as $-\vec{A}$. Addition of a vector and its negative vector results a vector of zero magnitude, which is known as a null vector. A null vector is denoted by arrowed zero $(\vec{0})$.

The idea of negative vector explains operation of subtraction as addition of negative vector. Accordingly to subtract a vector from another consider vectors $\vec{A}$ and $\vec{B}$ shown in the figure. To subtract $\vec{B}$ from $\vec{A}$, the negative vector $-\vec{B}$ is added to $\vec{A}$ according to the triangle law as shown in figure-II.



Multiplying by a Number

Multiplication by a positive number changes magnitude of the vector but not the direction and multiplication by a negative number changes magnitude and reverses direction.

Thus multiplying a vector by a number n makes magnitude of the vector n times. $n\vec{A}=(nA)\hat{a}$

Here $\hat{a}$ denotes the unit vector in the direction of vector $\vec{A}$.

Resolution of a Vector into Components

Following laws of vector addition, a vector can be represented as a sum of two (in two-dimensional space) or three (in three-dimensional space) vectors each along predetermined directions. These directions are called axes and parts of the original vector along these axes are called components of the vector.

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Cartesian Components in Two Dimensions

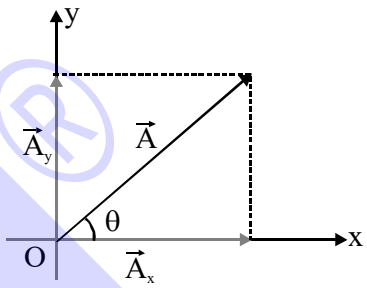
If a vector is resolved into its components along mutually perpendicular directions, the components are called Cartesian or rectangular components.

In figure is shown, a vector $\vec{A}$ resolved into its Cartesian components $\vec{A}_x$ and $\vec{A}_y$ along the x and y-axis. Magnitudes A_x and A_y of these components are given by the following equation.

$A_x = A\cos\theta$ and $A_y = A\sin\theta$

$\vec{A} = A_x\hat{i} + A_y\hat{j}$

$A = \sqrt{A_x^2 + A_y^2}$

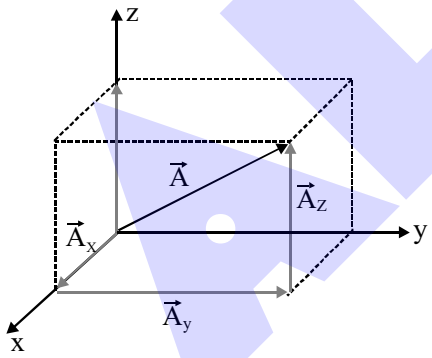


Here $\hat{i}$ and $\hat{j}$ are the unit vectors for x and y coordinates respectively.

Mathematical operations e.g. addition, subtraction, differentiation and integration can be performed independently on these components. This is why in most of the problems use of Cartesian components becomes desirable.

Cartesian components in three dimensions

A vector $\vec{A}$ resolved into its three Cartesian components one along each of the directions x, y, and z-axis is shown in the figure.



$\vec{A} = \vec{A}_x + \vec{A}_y + \vec{A}_z = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$; $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$

Equal Vectors

Two vectors of equal magnitudes and same directions are known as equal vectors. Their x, y and z components in the same coordinates system must be equal.

If two vectors $\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$ and $\vec{b} = b_x\hat{i} + b_y\hat{j} + b_z\hat{k}$ are equal vectors, we have

$\vec{a} = \vec{b} \Rightarrow a_x = b_x$, $a_y = b_y$ and $a_z = b_z$

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Parallel Vectors

Two parallel vectors must have the same direction and may have unequal magnitudes. Their x, y and z components in the same coordinate system bear the same ratio.

Consider two vectors $\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$ and $\vec{b} = b_x\hat{i} + b_y\hat{j} + b_z\hat{k}$, if they are parallel, we have

$$\Rightarrow \frac{a_x}{b_x} = \frac{a_y}{b_y} = \frac{a_z}{b_z}$$

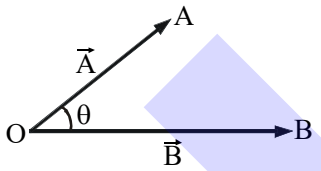
Product of Vectors

In all physical situation, whose description involve product of two vectors, only two categories are observed. One category where product is also a vector involves multiplication of magnitudes of two vectors and sine of the angle between them, while the other category where product is a scalar involves multiplication of magnitudes of two vectors and cosine of the angle between them. Accordingly, we define two kinds of product operation. The former category is known as vector or cross product and the latter category as scalar or dot product.

Scalar or Dot Product of Two Vectors

The scalar product of two vectors $\vec{A}$ and $\vec{B}$ equals to the product of their magnitudes and the cosine of the angle θ between them.

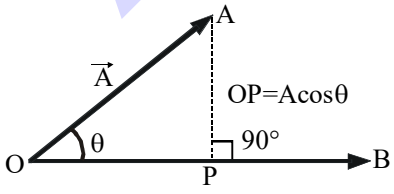
$$\vec{A}\cdot\vec{B} = AB\cos\theta = OA\cdot OB\cdot\cos\theta$$

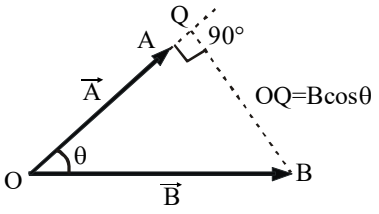


The above equation can also be written in the following ways.

$$\vec{A}\cdot\vec{B} = (A\cos\theta)B = OP\cdot OB$$

$$\vec{A}\cdot\vec{B} = A(B\cos\theta) = OA\cdot OQ$$





Above two equations and figures, suggest a scalar product as product of magnitude of the one vector and magnitude of the component of another vector in the direction of the former vector.

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